

# Does the stop decision reduce to the binding constraint?

Pit-loss cost and the tyre/fuel boundary across 205 endurance race-seasons

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## The question

The tank sets a floor on the number of stops; tyre degradation can in principle justify one more than that floor. My question is whether that is settled by the price of the binding constraint alone — once a stop costs enough, no car's degradation is steep enough to pay for it — rather than by the machinery. Your competitor-aware work prices charge and the stops it forces; this asks whether the price of the constraint, not the vehicle, sets the regime.

## How the two quantities are measured

Pit loss, per green-flag stop, in-lap and following out-lap both under green:

$$\text{loss} = t_{\text{in}} + t_{\text{out}} - 2 \times \text{that driver's median green pace in that race}$$

so it carries pit-lane transit and stationary time together, reported per class as a median.

Fuel range  $F$  is the ceiling of the 0.9 quantile of laps between pit visits, not the maximum, which is contaminated by cars that had stopped racing.

Tyre-limited, by exact dynamic program. For race length  $L$ , green pace  $p$ , fitted net slope  $\beta$  and pit loss  $\pi$ , a plan partitions  $L$  into stints ( $n_1 \dots n_{m+1}$ ) with  $m$  stops, costing

$$T = m \cdot \pi + \sum_j (n_j \cdot p + \beta \cdot n_j(n_j+1)/2) \quad \text{subject to } n_j \leq F$$

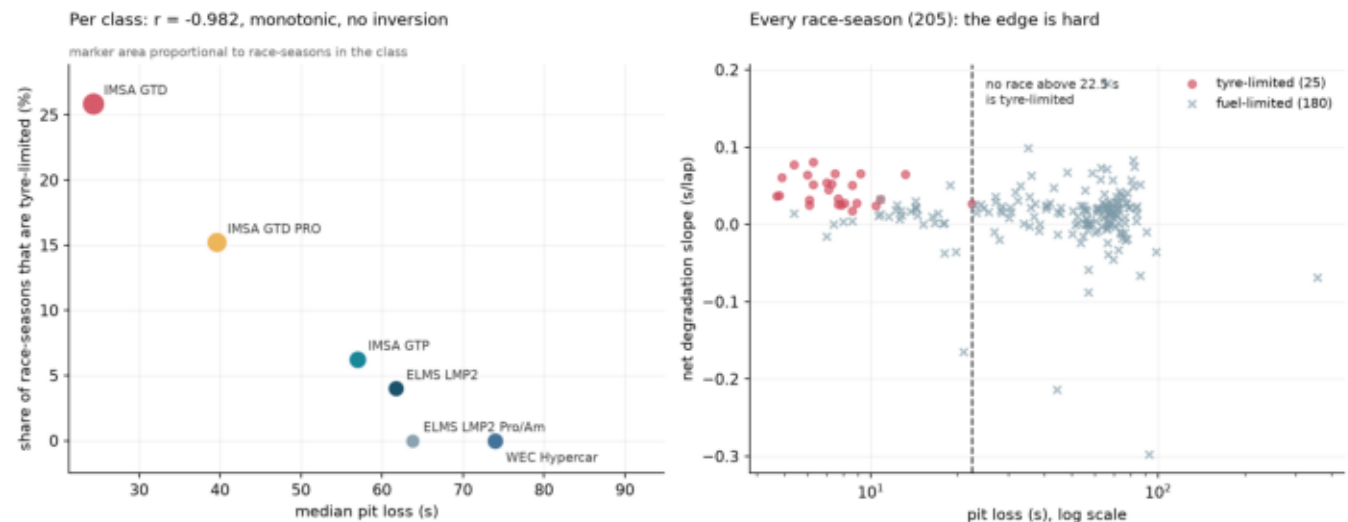
minimised exactly for each  $m$  from the fuel minimum  $\lfloor L/F \rfloor - 1$  upward; tyre-limited means the time-optimal  $m$  exceeds that minimum. The triangular sum is exact rather than the  $\beta \cdot n^2/2$  integral — they differ by  $\beta \cdot n/2$ , about a second over a forty-lap stint, and that is the only quantity weighed against  $\pi$ .  $\beta$  is fitted per race-season with driver-race fixed effects, as a net slope: the source carries no compound label.

## The correlation

| class            | median pit loss | tyre-limited | race-seasons |
|------------------|-----------------|--------------|--------------|
| IMSA GTD         | 24.4 s          | 25.9 %       | 58           |
| IMSA GTD PRO     | 39.6 s          | 15.2 %       | 46           |
| IMSA GTP         | 57.0 s          | 6.2 %        | 32           |
| ELMS LMP2        | 61.7 s          | 4.0 %        | 25           |
| ELMS LMP2 Pro/Am | 63.8 s          | 0.0 %        | 17           |
| WEC Hypercar     | 74.0 s          | 0.0 %        | 27           |

$r = -0.982$ , 95% CI  $[-0.986, -0.745]$ , monotonic across all six classes with no inversion. The interval is bootstrapped over races within class (205 races, 10,000 replicates), not over the six class summaries — six fixed championships are not a source of variation. At race level, 150 of 205 race-seasons sit above 22.5 s of pit loss and 0 of those is tyre-limited, across every class (25 of 205 are tyre-limited in total).

## The cost of the stop, not the car, decides whether tyres ever beat fuel



## Where this evidence is thin

The 22.5 s edge is a maximum, and one race puts it there: IMSA GTD at Indianapolis 2025. The next tyre-limited race sits at 13.2 s, so removing that one race moves it by 41%. I report no interval for it: a maximum sits on the boundary of its own support, so the bootstrap is degenerate above it and understates uncertainty below. The rule is not thin — the constant is.

## What I would value your view on

Whether a correlation across six class summaries is the right statistic at all. The alternative I see is a per-race model with pit loss as covariate and tyre-limited as outcome, using all 205 race-seasons instead of six points, at the cost of modelling the within-class dependence those summaries absorb.